DEPARTMENT OF STATISTICS

ST. XAVIER'S COLLEGE (AUTONOMOUS), KOLKATA

Name : Soham Choudhury

Roll No. : 21-300-4-07-0467

Registration No. : A01-1112-0876-21

Supervisor : Prof. Pallabi Ghosh

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Title : Time Series Analysis of Monthly Gold Price Data

DECLARATION

I affirm that I have identified all my sources and that no part of my dissertation paper uses unacknowledged materials.

Soham Choudhury

Soham Choudhury

Department of Statistics

St. Xavier's College (Autonomous), Kolkata

Date : April, 2024

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ABSTRACT

In this project we considered the data set of monthly gold price in USD. Here our target is to forecast the price of gold for some future time points. For this purpose we divided the dataset into two parts – training dataset and test dataset. At first we perform necessary tests and visualizations to understand the behaviour of gold price with respect to time. Then we fit an ARIMA model to the training dataset with appropriate parameters and forecast future gold price based on the fitted model and compare it with the test dataset.

INTRODUCTION

Historically, gold was used for supporting trade transactions around the world besides other modes of payment. Various states maintained and enhanced their gold reserves and were recognized as wealthy and progressive states. In present times, precious metals like gold are held with central banks of all countries to guarantee repayment of foreign debts, and also to control inflation. So gold price is a very important thing to be studied.

Data Description:

Source : https://datahub.io/AcckiyGerman/gold-prices

Title of the Dataset : Gold Prices (Monthly in USD)

The dataset contains monthly gold price in USD from January,1970 to July,2020.

Data Overview:

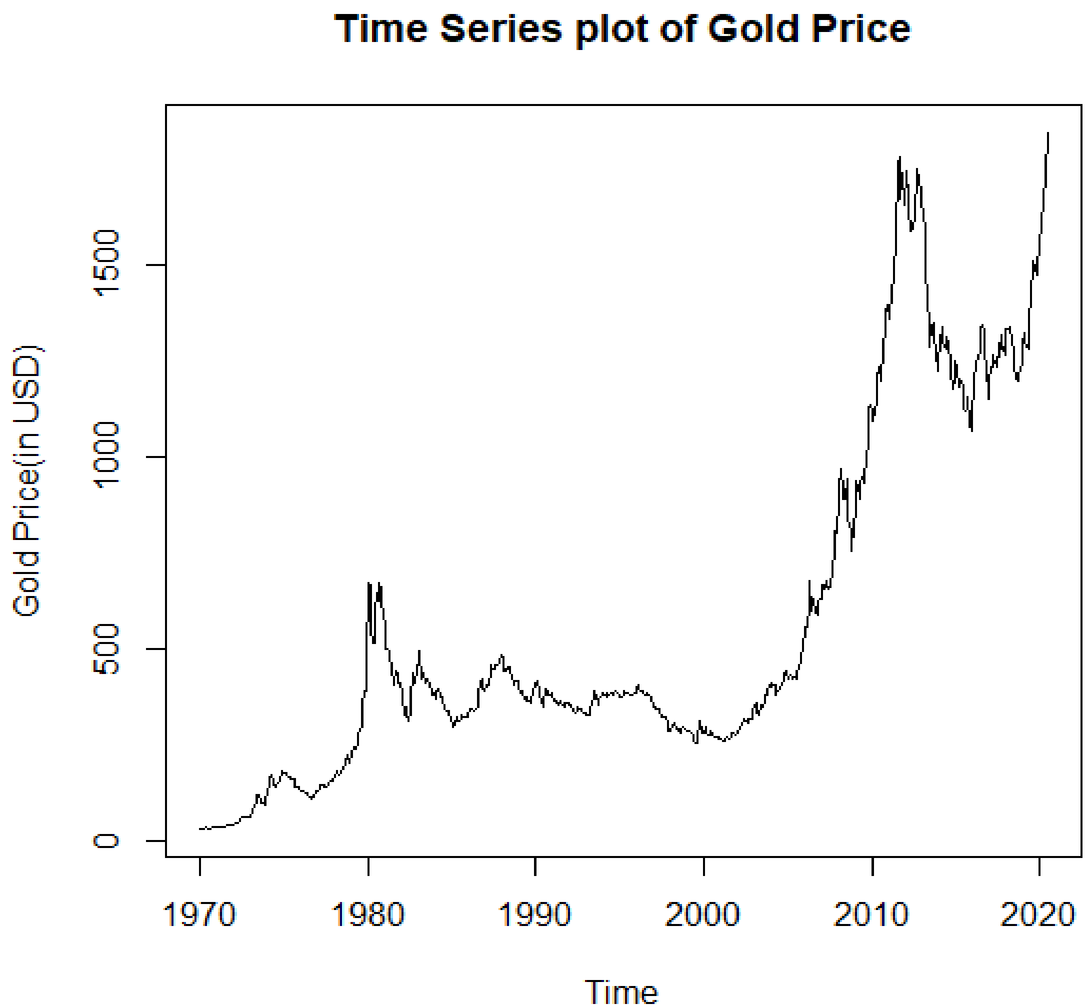
To have a primary understanding we tabulate the data of initial 5 years.

Time	Price (in USD)
1970-January	34.946
1970-February	34.994
1970-March	35.089
1970-April	35.623
1970-May	35.958
1970-June	35.437
1970-July	35.329
1970-August	35.377
1970-September	36.196
1970-October	37.553
1970-November	37.455
1970-December	37.434
1971-January	37.868
1971-February	38.716
1971-March	38.873
1971-April	39.001
1971-May	40.493
1971-June	40.105
1971-July	40.929
1971-August	42.722
1971-September	41.976
1971-October	42.473
1971-November	42.842
1971-December	43.455
1972-January	45.64
1972-February	48.237
1972-March	48.288
1972-April	49.026
1972-May	54.5
1972-June	62.17
1972-July	65.558
1972-August	66.917
1972-September	65.589
1972-October	64.824

1972-November	62.726
1972-December	63.779
1973-January	65.127
1973-February	73.971
1973-March	84.105
1973-April	90.441
1973-May	101.623
1973-June	119.8
1973-July	120.364
1973-August	106.225
1973-September	103.034
1973-October	99.923
1973-November	94.645
1973-December	106.236
1974-January	129.027
1974-February	150
1974-March	168.298
1974-April	172.243
1974-May	163.568
1974-June	154.013
1974-July	142.283
1974-August	154.362
1974-September	151.66
1974-October	158.533
1974-November	181.483
1974-December	183.683

Analysis:

We have monthly gold prices per ounce (Oz) since January, 1970 to July, 2020 in USD. To have an initial understanding about the dataset, first we plot the gold price data and observe its nature.



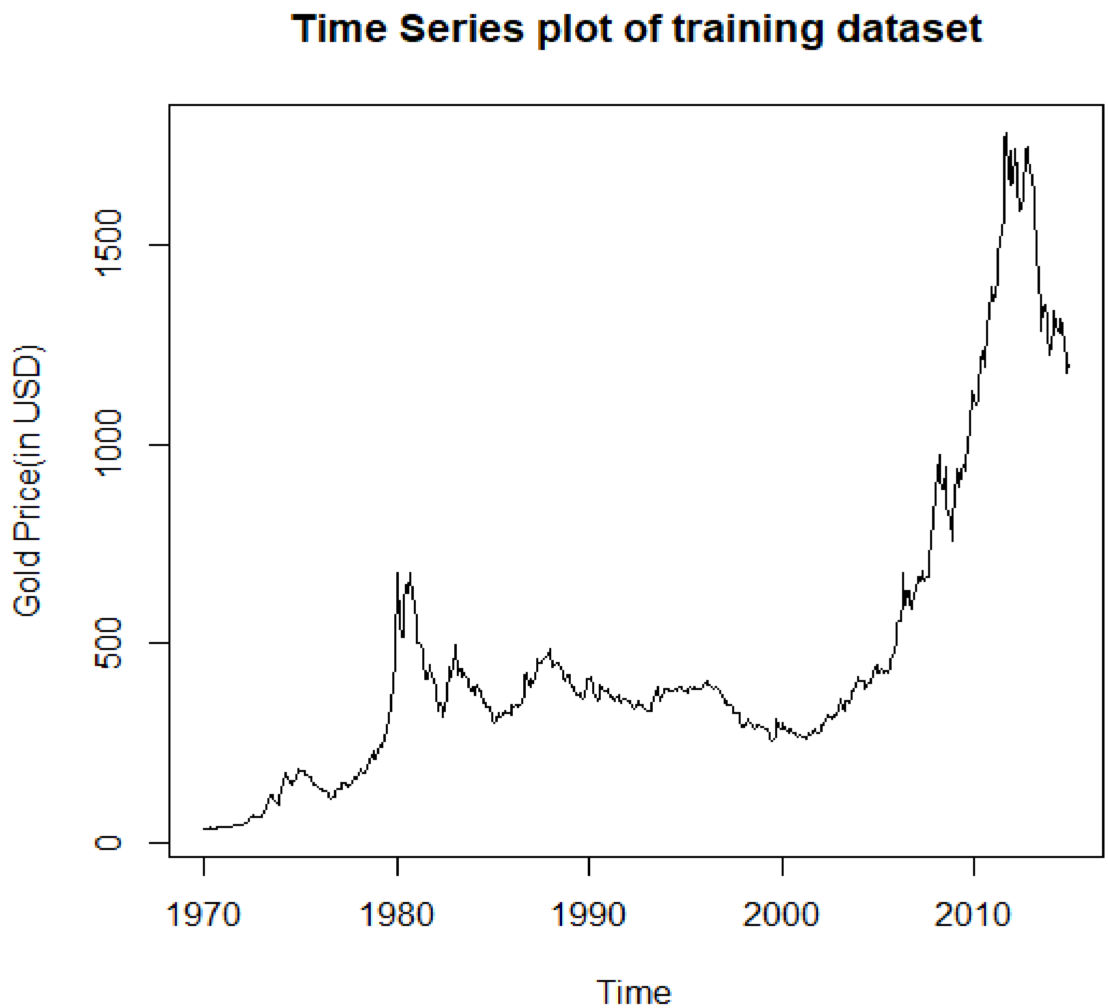
Data Preparation

Now we divide the dataset into two parts – training and test dataset. We fit an appropriate model to the training dataset and forecast future observations and compare it with the test dataset.

Training dataset : From Jan, 1970 to Dec, 2014 (No. of training data – 540)

Test dataset : From Jan, 2015 to Jul, 2020 (No. of test data – 67)

We plot the training data to understand the deterministic components present in the dataset.



Clearly from the graph we can see that trend component is a dominant factor present in the data. From the graph we get an idea that the data is non-stationary. To establish our assumption, we test for the stationarity of the training dataset.

Test for Stationarity : Augmented Dickey-Fuller Test

Dickey and Fuller devised a procedure to formally test for the presence of stationarity for a given dataset. The Augmented Dickey-Fuller test simply includes AR(p) terms of ΔX_t term in the following model. Therefore we have :

$$\Delta X_t = a + bt + \gamma X_{t-1} + \sum_{i=1}^{p-1} \beta_i \, \Delta X_{t-1} + e_t$$

Where a is a suitable constant, b is the coefficient on a time trend.

The Augmented Dickey-Fuller statistic, used in this test, is a negative number. The more negative it is, the stronger the rejection of the hypothesis that there is a unit root at some level of confidence. Here the null hypothesis is, $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$. The test statistic is given by,

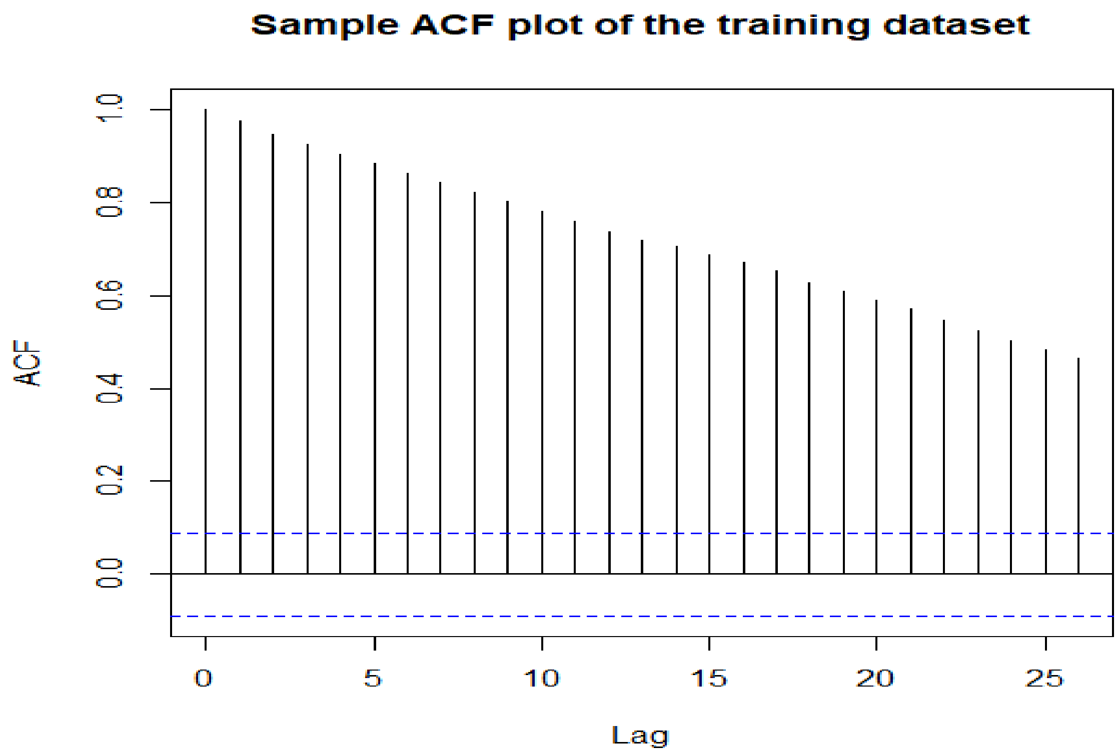
$$\frac{\hat{\gamma}}{SE(\hat{\gamma})}$$

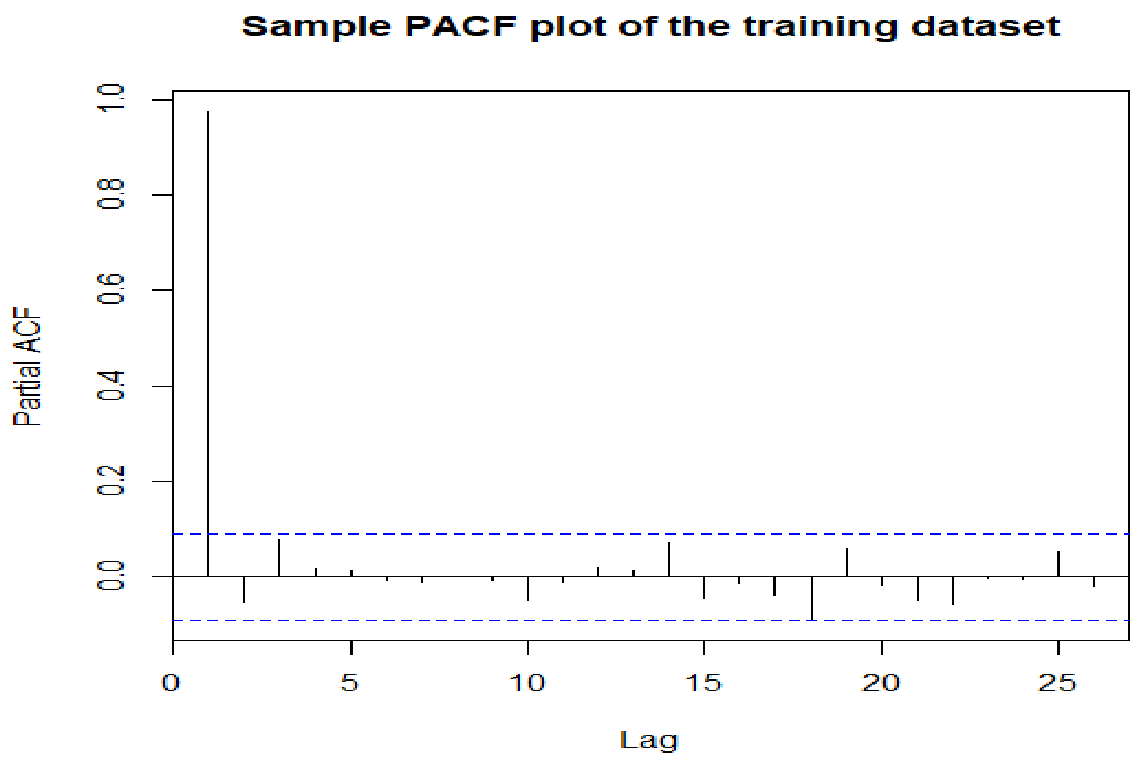
Here the p-value of the test is 0.7621. So we shall accept H_0 at 5% level of significance i.e. In the light of the given data we conclude that the data is non-stationary.

Model Identification

Firstly, we shall plot the sample ACF and PACF of the training dataset for model identification purpose. Then we shall estimate the model parameters appropriately.

ACF and PACF plots



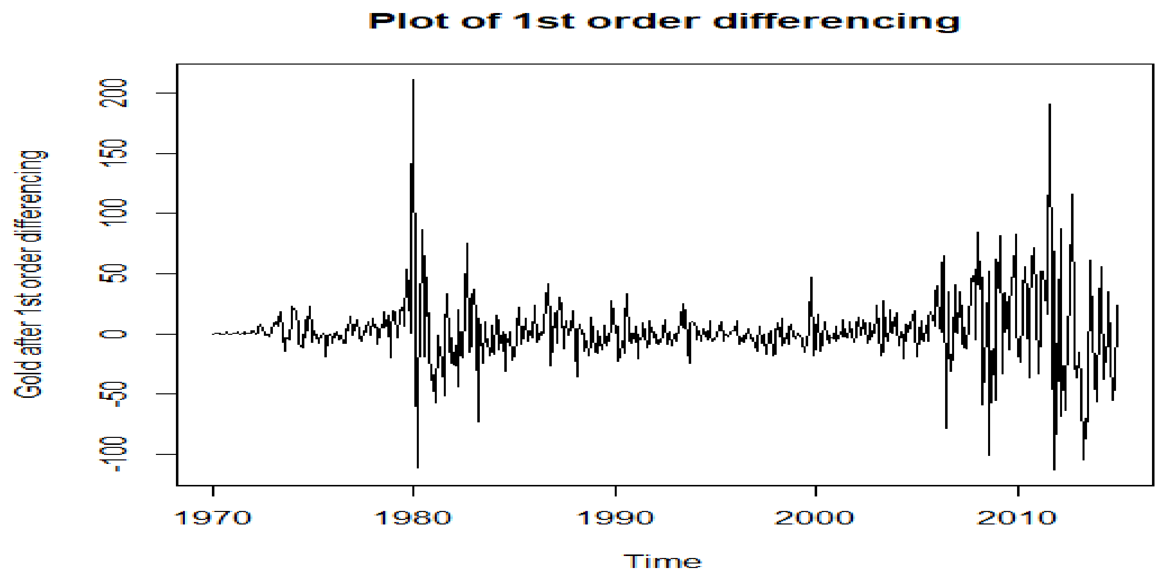


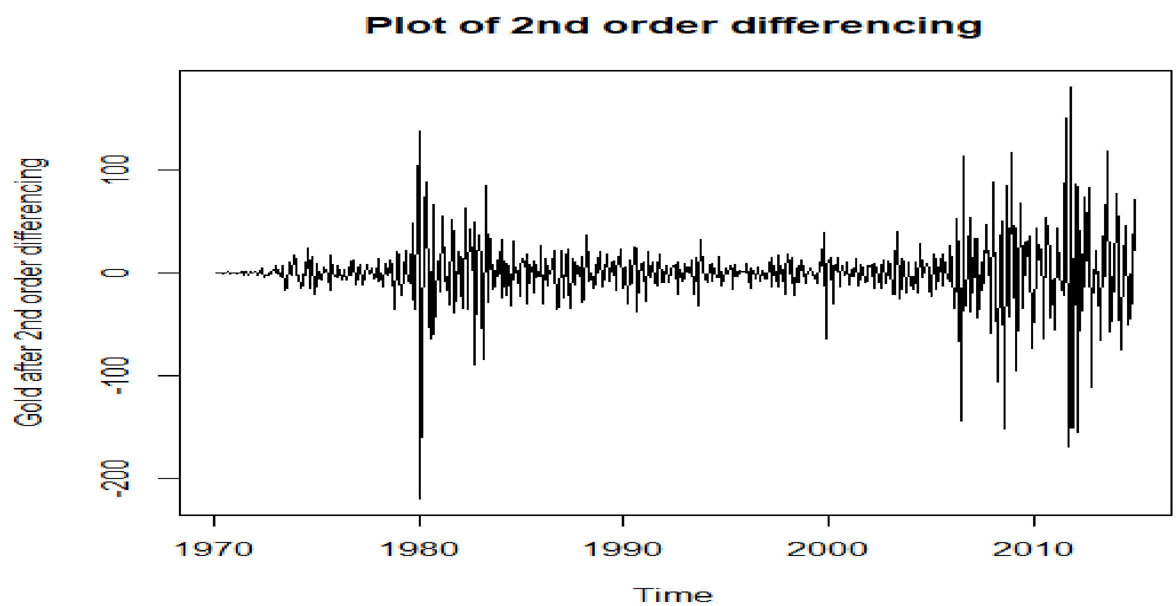
We previously tested that our data is non-stationary. The sample ACF plot is slowly decaying with positive autocorrelation. This is a distinctive feature to suggest the appropriateness of an ARIMA model. Hence, from the above plots we can say that an ARIMA model could be appropriate in this case.

ARIMA model uses previous historic data and decomposes it. Here Autoregressive (AR) indicates weighted moving average over past data, Integrated (I) indicates linear trend or polynomial trend, Moving Average (MA) indicates weighted moving average over past errors.

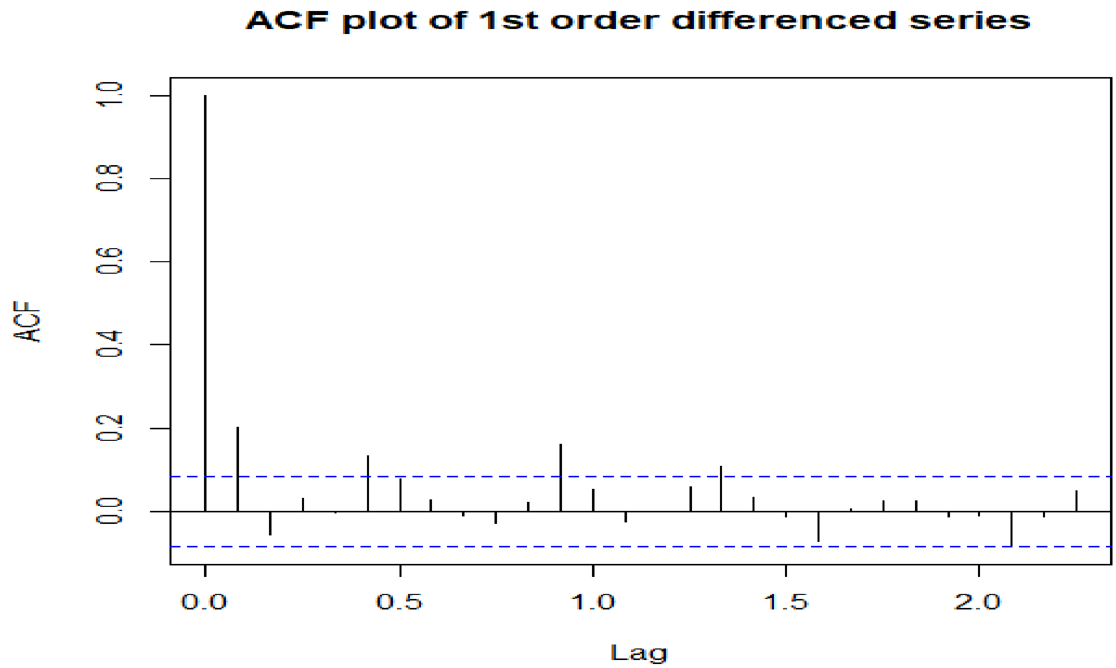
Estimation of ARIMA model parameters

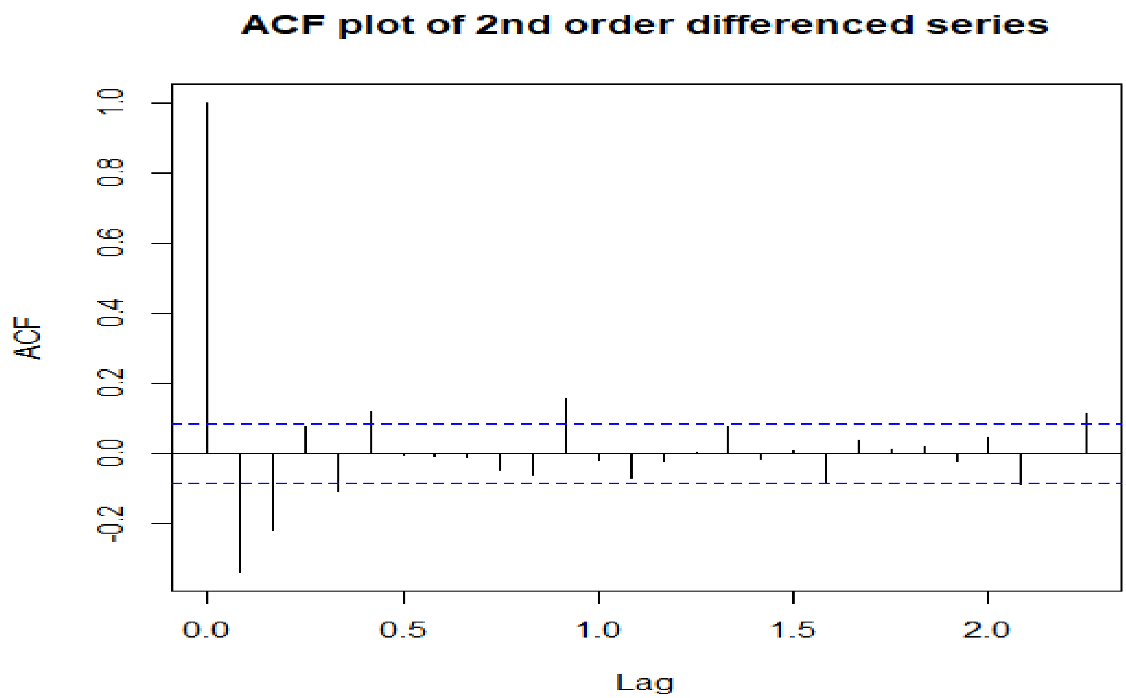
From the sample ACF and PACF, the p value can be chosen as 1 for ARIMA(p,d,q) model since sample PACF cuts after lag 1 i.e. PACF plot has a significant spike only at lag 1. To decide about the parameter d, we difference the prices for difference value 1 and 2. Now, we plot the differenced series.





From the graphs, the differenced series of order 1 and 2 seems to be stationary. We perform Augmented Dickey-Fuller test to the differenced series to check whether it is stationary or not. Here the p-value is less than 0.01 for both the cases. Hence, we conclude that the 1^{st} and 2^{nd} order differenced series are stationary. We plot the sample ACFs of the differenced series.





From the ACFs, we can see that in 2nd order differencing the immediate lag is negative implying that the series has become over differenced. So we choose the value of the parameter d as 1.

The ACF of the original data does not help us to choose the value of the parameter q. From the ACFs of the differenced series, we get an idea that the value of q can be chosen as 1 or 2.

We fit different ARIMA(p,d,q) models for all the combinations of p,d,q assumed from the ACF, PACF plots and compare the error measures to choose the best model.

Here the measures taken into consideration are,

i) Root Mean Square Error (RMSE) : Let h_t be a set of m number of observations varying with time. Define $\overline{h}_t = \frac{1}{m}\sum_{t=1}^{m} h_t$. Then RMSE is given by $\sqrt{\sum_{t=1}^{m}\frac{(ht-h\bar{t})^2}{m}}$

ii) Mean Absolute Percentage Error (MAPE) : MAPE is given by $\frac{1}{m}\sum_{t=1}^{m}\frac{|ht-h\bar{t}|}{|ht|} \times 100\%$

iii) Mean Absolute Error (MAE) : MAE is given by $\frac{1}{m}\sum_{t=1}^{m}|ht-h\bar{t}|$

iv) Akaike's Information Criterion (AIC) : AIC can be written as

$$AIC = 2k - 2(\text{log-likelihood})$$

Where k is the number of parameters estimated in the model.

Table-1.1

ARIMA(p,d,q)	RMSE	MAPE	MAE	AIC
ARIMA(1,1,1)	26.72027	3.274712	15.71238	5078.48
ARIMA(1,1,2)	26.68538	3.270287	15.71029	5079.06
ARIMA(1,2,2)	26.73095	3.264083	15.66285	5079.13
ARIMA(1,2,1)	27.05876	3.265029	15.75066	5089.31

From the table we can clearly see that ARIMA(1,1,1) model has relatively low RMSE, MAPE, MAE and the least AIC. So based on AIC, ARIMA(1,1,1) is the best model to fit.

Fitting of ARIMA(1,1,1) Model

We fit ARIMA(1,1,1) model to our data and plot the fitted values with the original values to check how well our model is fitted.

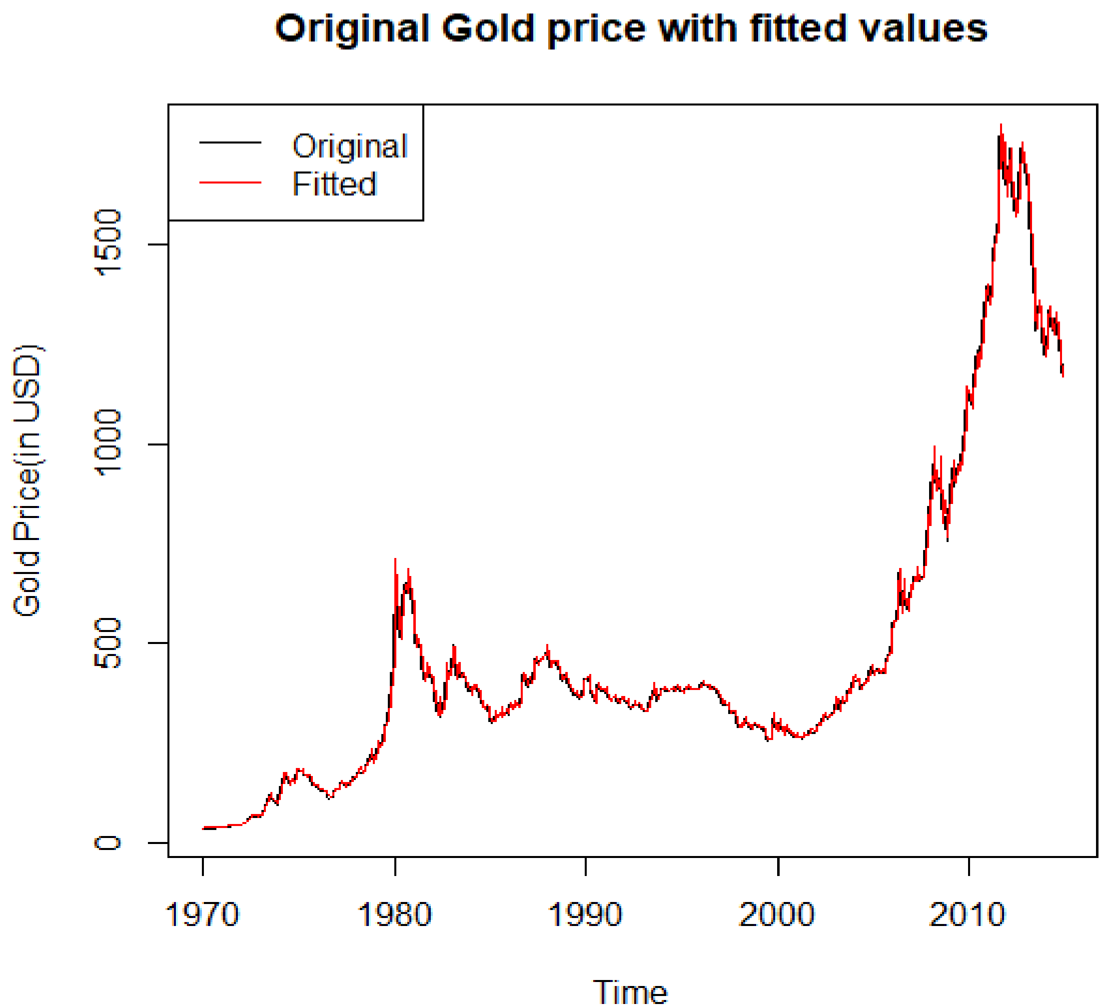
Thus the generalized model is given by,

$$X_t = \mu + \varphi_1 X_{t-1} + e_t - \theta_1 e_{t-1}$$

Where μ = process mean, φ_1 = AR coefficient, θ_1 = MA coefficient

For our dataset, the estimated values are,

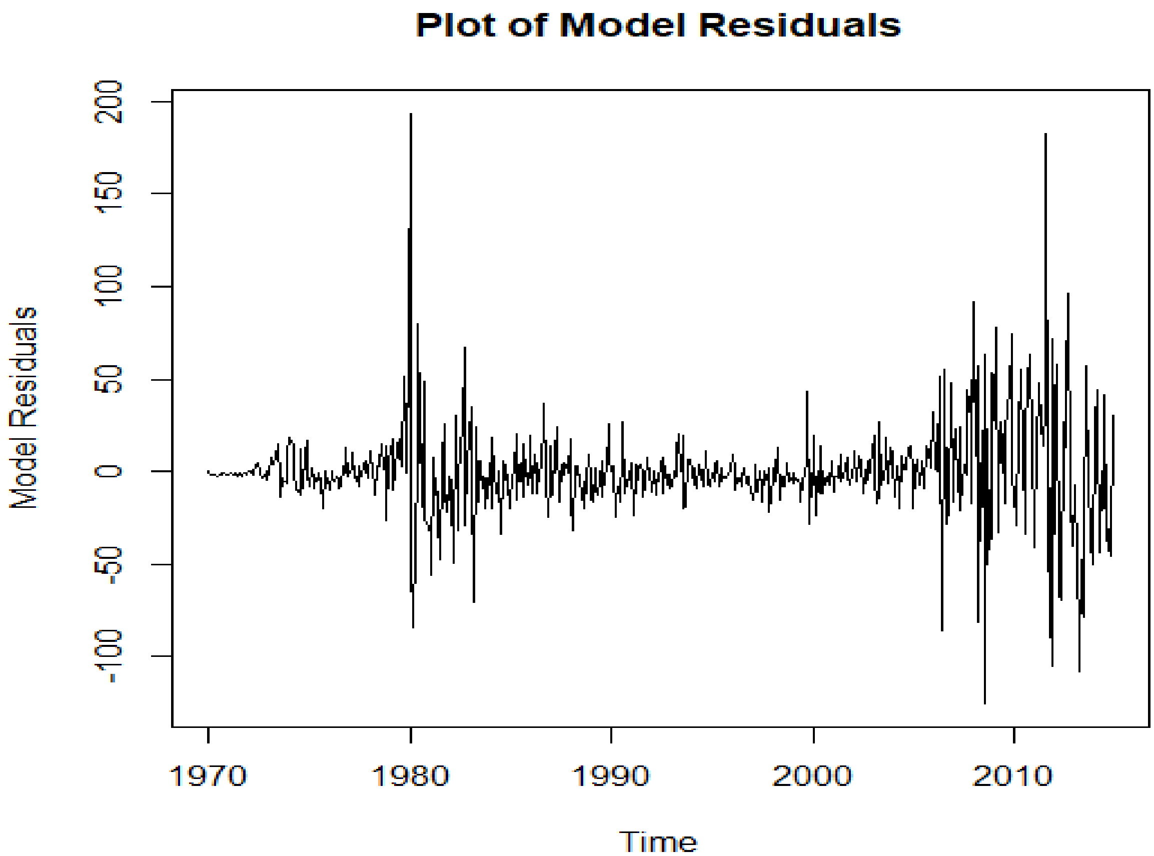
$$\hat{\mu} = 2.1730,\ \hat{\varphi}_1 = -0.5595,\ \hat{\theta}_1 = 0.7847$$



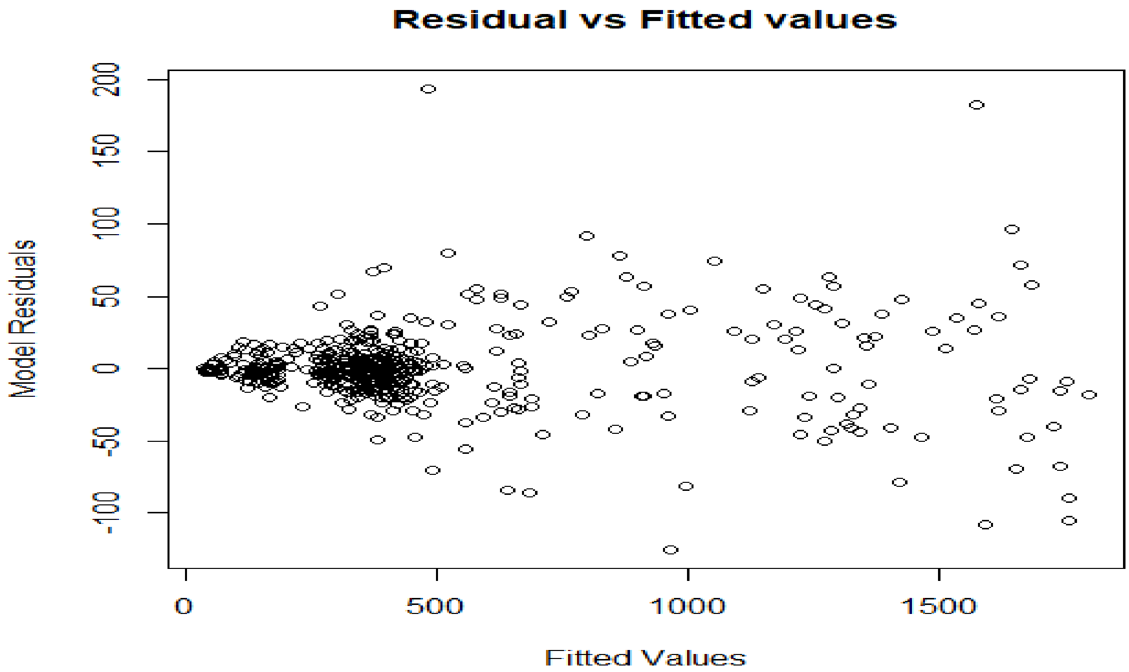
From the graph we can say that the fit is very good. Now we must check how the model residuals are behaving.

Residual Analysis

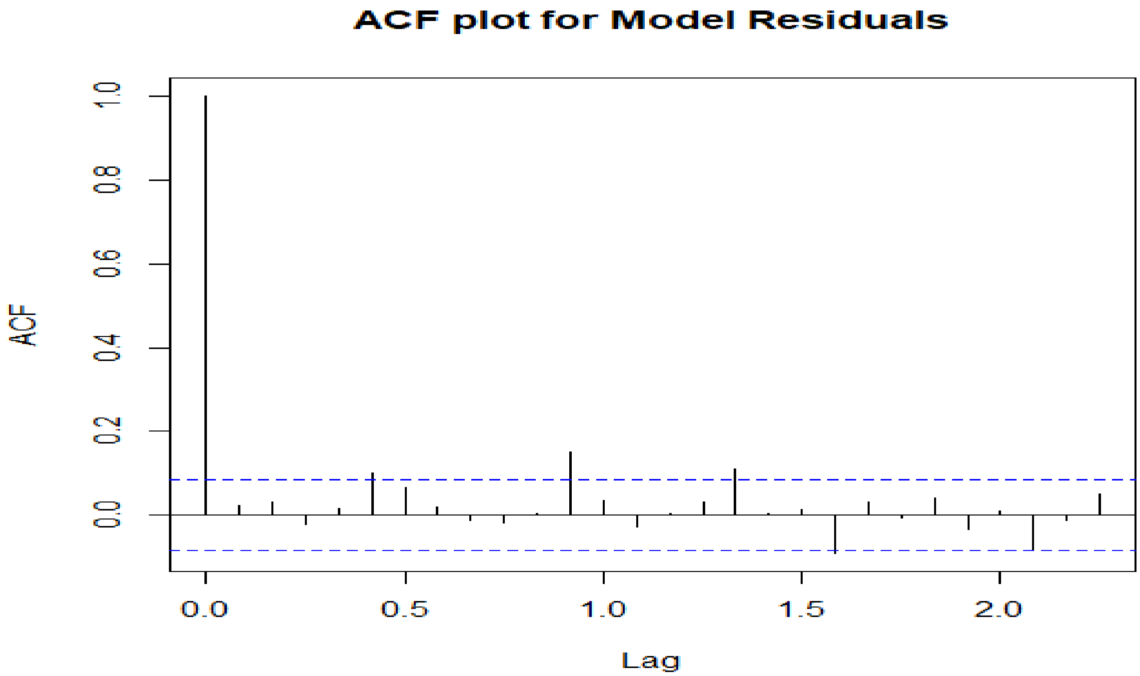
After fitting the ARIMA model, we must check whether the residuals are random or not. The plot of the model residuals is given below.



Now we shall look at the plot of model residuals against the fitted values. A random pattern indicates absence of any deterministic component.



From the above plot we can see that the pattern is random and hence, we can say there is no deterministic component present. Now, we look at the ACF plot of the model residuals.



From the ACF plot we can see only a single spike at lag 0 indicating purely randomness of the model residuals.

Now to establish our assumptions we can do the Run Test which will test whether the model residuals are random or not.

Test for Randomness : Run Test

This is a non-parametric test to detect randomness of a given set of observations. Here we are to test,

H_0 : The observations arise from a random process vs H_1 : The observations arise from a non-random process

A run is defined as uninterrupted sequence of one type of symbol. Let there be an ordered sequence of 'n' elements. There are n_1 elements of first type and n_2 elements of second type such that $n_1 + n_2 = n$. We assign '-' sign to the n_1 elements of first type and '+' sign to the n_2 elements of second type. The test statistic is given by R = Total number of runs.

For our test R = 257, i.e. R is odd

Hence, under the null hypothesis

$E(R|H_0) = \frac{2n1n2}{n1+n2} + 1$, $Var(R|H_0) = \frac{2n1n2(2n1n2-n1-n2)}{(n1+n2)^2(n1+n2-1)}$

For large sample, under H_0 $\frac{R-E(R|H0)}{\sqrt{Var(R|H0)}} \sim N(0,1)$

Here the p-value of the test is 0.2278. Hence we accept H_0 at 5% level of significance and conclude that the model residuals are random.

Forecasting

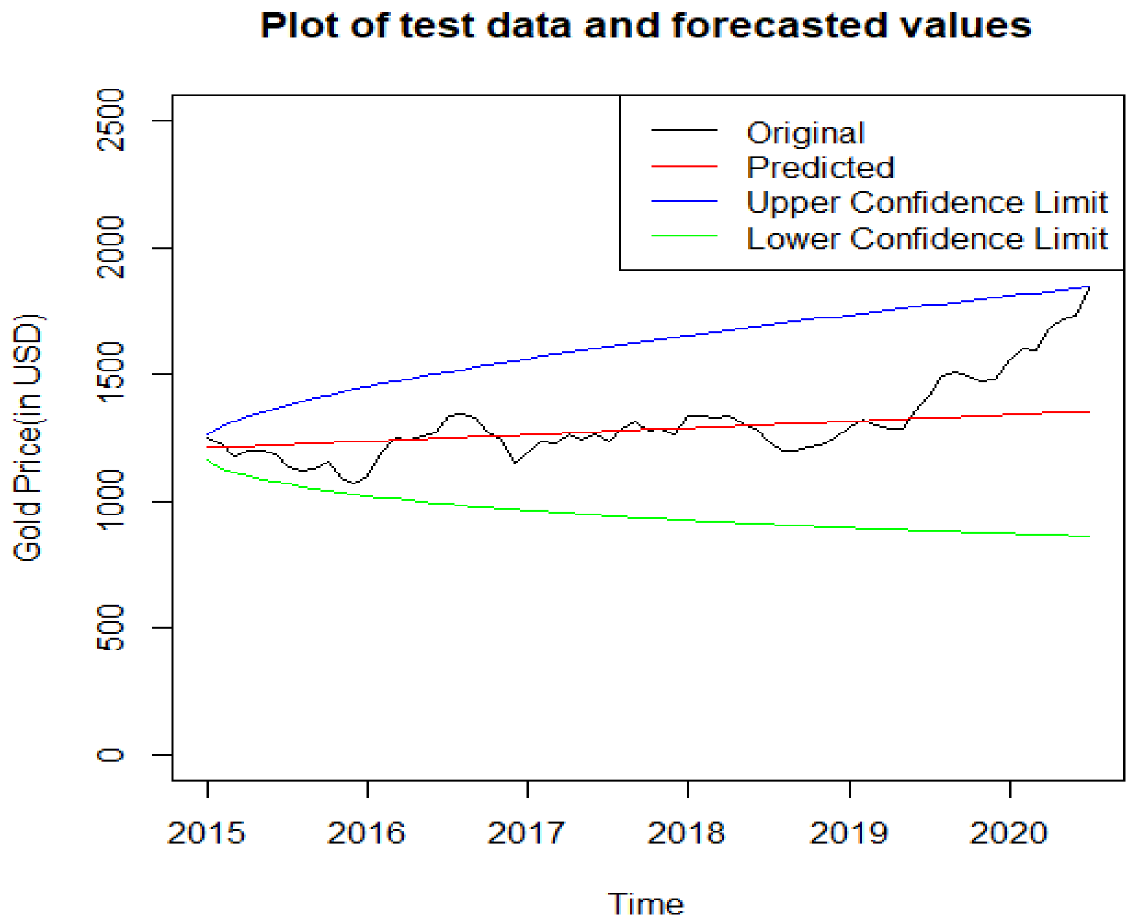
For the purpose of testing our fitted model, we left some observations of our dataset as test dataset. Now we shall forecast the gold prices for the test dataset and compare them with the actual values. To have a better understanding about the efficiency of fitting for the test dataset we look at the line plot of the original and predicted gold prices against the time points.

The forecast function is given by,

$$\widehat{X}_t = \hat{\mu} + \widehat{\varphi}_1 X_{t-1} - \widehat{\theta}_1 \varepsilon_{t-1}$$

(The estimates are obtained during the fitting of the model)

We generate the forecasted values using the forecast function in R.



The forecast function in R also provides us with the 95% confidence interval for the forecasted values. We can clearly see that all the datapoints in the test datasets lies within the confidence limits. Now to compare the forecasted values with actual values we use the measure Mean Absolute Percentage Error (MAPE). Here MAPE = 6.136842. Hence we conclude that the forecasted values are quite close to actual values though there are some fluctuations in some cases.

Conclusion

We can see that for predicting gold price in a short term, ARIMA model is working quite well. Although there are some limitations of ARIMA modeling. This is used only for short term forecasting and this model does not work well for long term. Also if there are too many variations, sudden changes in the data, ARIMA model fails to catch that. Hence this model becomes ineffective for long term forecasting and time series data with sudden changes in it.

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